

Foundations of Mycology & Anti-fungal Agents - Factoids

Week 8 • Core Lecture 7 • Fisher + Anderson co-primary

Official objective map: fungi vs bacteria • hyphae • sexual vs asexual reproduction • sexual vs asexual spores • antifungal MOA • major toxicities.

Five Fungi-vs-Bacteria Differences

Fungi	Bacteria
Eukaryotic: nucleus, mitochondria, ER	Prokaryotic
Wall = glucans + chitin	Wall = peptidoglycan
Membrane contains ergosterol/sterols	Lecture comparison: no sterols
Lecture: rare/no anaerobes	Many anaerobes
Some thermal dimorphism	No thermal dimorphism

Morphology

Term	Recognition
Yeast	Single budding cells
Mould	Filamentous hyphae; mycelium = mass of hyphae
Dimorphic	Deck expects yeast at 37 C; handout mnemonic = mould cold / yeast heat
Coenocytic	Aseptate; Zygomycete course pattern
Septate	Cross-walls present; Aspergillus pattern
Pseudohyphae	Chains of elongated attached budding yeast cells

Reproduction & Spores

Asexual / anamorph Favorable conditions • mitosis • most common in deck figure • blastoconidia/fission/ballistoconidia shown.		Sexual / teleomorph Unfavorable conditions • plasmogamy → karyogamy → diploid stage → meiosis → sexual spores.
Course group	Key spore / morphology	Example
Zygomycetes	Sporangiospores asexual; zygosporangium sexual; coenocytic	Mucor, Rhizopus
Basidiomycetes	Basidium → basidiospores	Cryptococcus, Malassezia
Ascomycetes	Conidia asexual; ascus → ascospores sexual; septate	Aspergillus

Extra Morphology Terms

Term	Lecture meaning
Mycelium	Mass of hyphal strands/branches with spore-bearing structures.
Thallus	Actively growing part of the hypha.
Heterothallic	Requires compatible mating strains for sexual reproduction.
Homothallic	Self-compatible.
Simple vs complex septa	Simple = solid plate/no pores; complex septa allow cytoplasm/organelles through pores.

Antifungal Target + Toxicity Map

Class/drug	Target / MOA	Major lecture toxicity / clue
Amphotericin B	Binds ergosterol → membrane pores	Nephrotoxicity ; K/Mg/Ca wasting; anemia; infusion chills/fever/headache
Azoles	Inhibit CYP 14α-demethylase → ↓ ergosterol	Hepatotoxicity • CYP interactions • GI • QTc prolongation
Terbinafine	Inhibits squalene epoxidase	Dermatologic/tinea use emphasized
Echinocandins	Fks / β-1,3-D-glucan synthase inhibition	Cell-wall class; Candida emphasis; poor urine/brain/eye penetration in summary
Flucytosine	Fungal conversion to 5-FU → inhibits RNA/protein + DNA synthesis	Bone marrow suppression ; hepatitis; oral; combination use

Azole Individual Clues

Drug	Remember
Fluconazole	Candida, Cryptococcus, tinea, dimorphic fungi
Voriconazole	Aspergillus; visual color changes; hallucinations/nightmares; photosensitivity; TDM
Posaconazole	Mucorales coverage; take oral form with food; prophylaxis; TDM; electrolyte issues
Itraconazole	Dimorphic fungi / onychomycosis; capsule needs acid; solution empty stomach; TDM
Isavuconazole	Aspergillus/mucormycosis; shortens QTc ; reversible LFT elevation

Spectrum / Practical Clues from Anderson Deck

Agent	Coverage / practical clue
Amphotericin B	Broad: Candida, Aspergillus, Mucor, dimorphic fungi, Cryptococcus, other molds; lipid formulations shown with improved CNS penetration.
Fluconazole	Candida, Cryptococcus, tinea, dimorphic fungi.
Voriconazole	Aspergillus + Candida + Scedosporium/Fusarium/dematiaceous molds; TDM.
Posaconazole	Candida, Aspergillus, dimorphic fungi, Mucorales; primarily prophylaxis; take oral form with food; TDM.
Itraconazole	Dimorphic fungi, aspergillosis, onychomycosis; capsule needs acid; solution empty stomach; TDM.
Isavuconazole	Aspergillosis + mucormycosis; QTc shortening distinguishes it.
Echinocandins	Good Candida; Aspergillus marked *; poor/no Mucor, Cryptococcus, Blastomycosis.

Newer / Lower-Priority Mechanisms

Agent	MOA
Ibrexafungerp	β-1,3-D-glucan synthesis, different Fks binding site; oral; less impacted by FKS mutation
Olorofim	DHODH → pyrimidine biosynthesis
Fosmanogepix	Gwt1 → anchored fungal adhesion proteins

Deck/table priority note: selected newer-agent details are marked informational/clinical-rotation material. Prioritize classic classes + toxicities for the official objectives.

Final 20 seconds: Ampho = ergosterol pores + kidney • azoles = 14α-demethylase + liver/CYP/QT • echinocandins = glucan wall • flucytosine = 5-FU + marrow • terbinafine = squalene epoxidase.

Primary: Fisher mycology deck + Anderson antifungal deck. Uploaded 10A/10B handouts are enrichment and retain older handout numbering only.